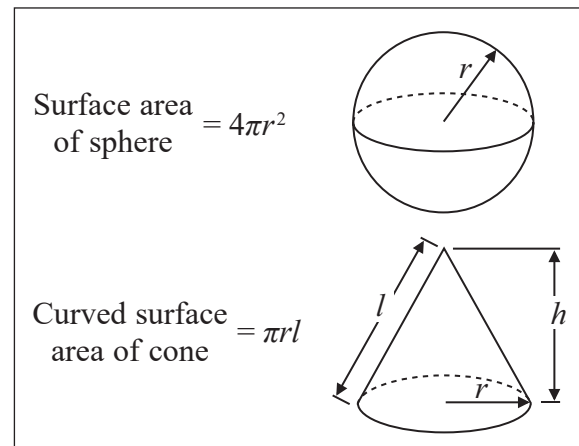
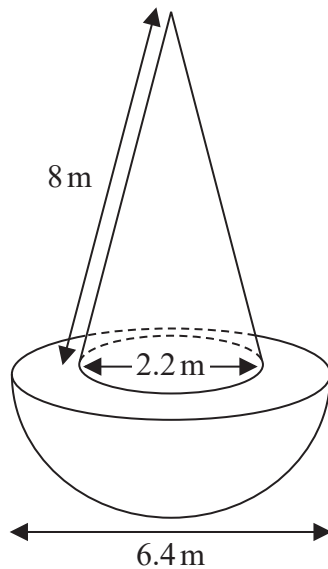


- 18 The centre of the base of a solid cone is placed at the centre of the flat face of a hemisphere to make a sculpture.



The diameter of the hemisphere is 6.4 m.

The diameter of the base of the cone is 2.2 m and the slant height of the cone is 8 m.

Callum is going to cover this sculpture with one coat of paint.

He assumes that each tin of paint will cover 8.5 m^2

Callum thinks he will need to buy 15 tins of paint.

(a) Will 15 tins of paint be enough?

You must show how you get your answer.

Callum finds out that each tin of paint will actually cover 7.5 m^2

(b) How does this affect your answer to part (a)?

(1)

(Total for Question 18 is 6 marks)